

Wavelet transform based approach for EEG feature selection of motor imagery data for brain-computer interfaces

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Abstract — The Electroencephalogram (EEG) signals recorded during motor imagery (MI) tasks are often used as input in the brain-computer interface (BCI) applications. The adaptation of these EEG signals to control signals depend on the selection of an accurate feature that can provide good classification. The study compared various features of alpha frequency band adapting wavelet-based time-frequency analysis approach. The mean, variance, wavelet energy, Shannon entropy, log energy entropy, kurtosis, and skewness feature were extracted from the EEG signals recorded from symmetrical electrode locations of the motor cortex and their performance accuracies were evaluated using SVM and KNN classification algorithms. In results, the Shannon entropy showed the highest classification accuracy of 86.4% with SVM classifier. Findings of this study establish Shannon entropy as a potential feature to use in EEG-based BCI analysis along with SVM classification algorithm.

Keywords— *Electroencephalogram (EEG), Feature selection, Motor Imagery (MI), Discrete Wavelet Transform (DWT), Support Vector Machine (SVM), K-Nearest Neighbor (KNN).*

I. INTRODUCTION

Brain-computer interface (BCI) defines a direct communication between the human brain and a computer to operate an external device [1]. The common idea to develop a BCI system was to help physically challenged but cognitively intact people to sustain at their own by controlling a machine with the brain signals [3]. The system can restore the motor ability or can establish a way of communication for an impaired individual by decoding the neural activation pattern. For the purpose, the BCI should be capable of transferring the neural information into commands that would control the external tool, such as a wheelchair, a robotic arm, a prosthetic limb or a computer cursor [6]. The technology has potential applications in automobile, robotics and in medical sciences [3].

Though various techniques such as electroencephalography (EEG), Electrocorticography (ECoG), functional magnetic resonance imaging (fMRI), and magnetoencephalography (MEG) are available for measuring neural activity, BCI applications have found EEG to be the most reliable non-invasive tool [1],[3]. Analysis of EEG signals associated with imagined finger or limb movement, commonly known as motor imagery (MI) and executing the actual physical movement using an artificially intelligent system forms the foundation of BCI [1]. Motor tasks generate event-related potentials in the

motor cortex which is located in the central region of the brain. In research applications, this primarily includes classifying left versus right imagined motor movements. EEG signals are nonstationary in nature and have a very high temporal resolution [3]. This oscillatory signal varies from 0.5 to 100 μ V in amplitude and it has different frequency components in delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz) and gamma (<30 Hz) bands [5]. Previous MI related BCI studies have suggested that the changes in alpha and beta frequency bands originating at sensory motor cortex efficiently represents the MI-induced neural activation [1],[3]. The task-induced event-related potentials, which are event-related synchronization (ERS) and desynchronization (ERD) pattern of EEG in these two frequency bands, specifically in alpha, were been extensively studied in the literature [1],[6].

Both time and frequency information content play a dominant role in EEG analysis. Since the frequency information changes over time, the time-frequency based wavelet domain provides a better representation of complex EEG signals than the Fourier domain [11]. Wavelet transform can maintain the time as well as frequency resolution of the signals leading to effective feature extraction that affect the classification accuracy of the BCI.

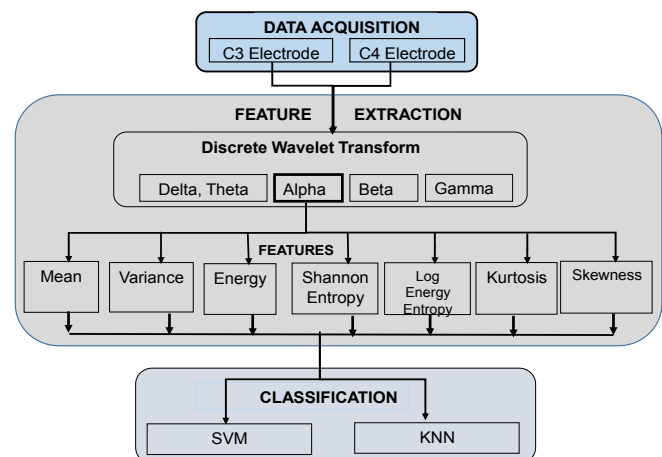


Fig. 1. Block diagram representation of the current work

The features are the statistical measure of a time domain signal that can represent the signal characteristics. Studies have shown that appropriate feature can lead to a good classification accuracy even if an unsophisticated classifier is adopted [1],[2]. Some common wavelet-based features

used in MI classification includes wavelet energy [12],[13], entropy [14], mean [1],[13],[9], standard deviation [9], variance [1], band power [6], wavelet coefficients [15] etc.

The objective of this paper was to identify the most reliable and efficient EEG feature for MI classification. Fig.1 displays the block diagram of the algorithm used in this study. A brief introduction to the objective of the study has been explained in the introduction section. In section II, the methodology of the study is explained. It describes the data collection, feature extraction and classification algorithms employed in the study. Section III discusses the results obtained and section IV concludes the analysis.

II. Methodology

A. Data Collection

The study incorporated the datasets of BCI 2003 competition which is known also as Graz2003 data. This dataset has been used as a benchmark in several motor imagery classification problems. This dataset was recorded by the Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology Graz from a healthy female participant (age = 25yrs) while imagining a left and right-hand movement [13],[9]. The task was to move a block to either of the ways by the imagery EEG signals where the order of left cues and right cues are randomly placed. During the task performance, the participant was made to relax on a relaxing chair having armrests. There were 7 sessions each of 40 trials. In each trial, at the 2nd second, a sound signal appears along with a '+' sign on the screen indicating the start of the trial. At the 3rd second, the visual trials appear on the screen displaying left or right directive arrows (Fig.2). There are a total of 280 trials, each with 9-second duration in the dataset. 140 trials are for training and 140 are for testing.

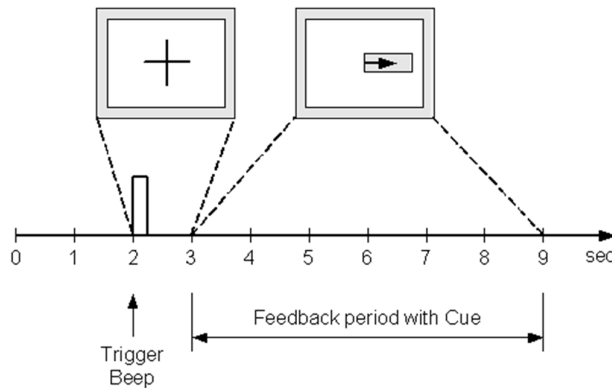


Fig.2. Timing scheme of the task

The EEG signals were recorded from the participant while showing the cues and the BCI needs to feedback the data to move the arrow. The dataset considered the electrodes of C3, Cz and C4 (Fig.3) which are capable of recording brain potentials for motor imagery out of the motor cortex. We have taken only C3 and C4 electrodes for

having most relevance for left and right-hand imagery [3]. The sampling rate was 128 Hz.

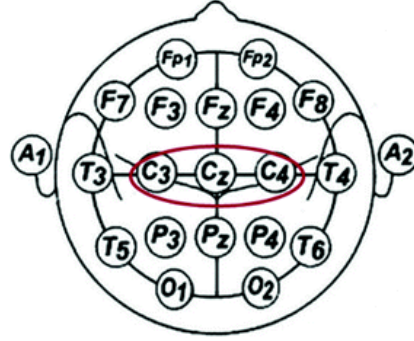


Fig.3.

Electrode locations used for data acquisition

B. Feature Extraction

Accurate analysis of EEG signals is dependent on extracting the accurate information content. For feature extraction, first, the alpha frequency band is extracted from the filtered EEG signal using Discrete Wavelet Transform (DWT) [9]. For any j^{th} level, the DWT algorithm decomposes a signal into low and high frequency coefficients, called approximate coefficient, A_j and detail coefficient D_j respectively as given by Eqn. (2) and (3).

$$A_j[n] = \sum_i x[i] g[2n - i] \quad (2)$$

$$D_j[n] = \sum_i x[i] h[2n - i] \quad (2)$$

Where $g[n]$ represents the low pass filter and $h[n]$ represents the high pass filter.

In the next successive level, the approximate coefficient is further decomposed and the algorithm continues (Fig.4). For p-levels, the DWT produces one approximate coefficient and p number of detail coefficients.

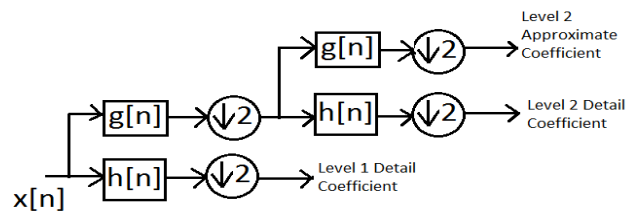


Fig. 4. Discrete Wavelet Transform scheme of level 2.

In this study, the maximum frequency present in each signal was 64Hz (Half of the sampling rate), hence the details coefficient of 3rd level generated alpha frequency band (8-16 Hz). From each filtered signal collected from C3 and C4 electrode locations, the alpha band was extracted and forwarded for the feature extraction process. The study calculated the following statistical feature to compare.

- (a) Mean (μ): The mean of a signal $x[n]$ is given by,

$$\mu = \frac{1}{N} \sum_{i=1}^N x[i] \quad (4)$$

- (b) Variance (σ^2): The variance of a signal $x[n]$ is given by,

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x[i] - \mu)^2 \quad (5)$$

The parameter σ indicates the standard deviation of the finite signal $x[n]$.

- (c) Energy (En): Energy of a signal $x[n]$ is given by

$$E = \sum_{i=1}^N |x[i]|^2 \quad (6)$$

The normalized energy of the alpha frequency band can be computed as a ratio of alpha band energy (En_{alpha}) to the energy of the whole signal, En_{total} .

$$En = \frac{En_{alpha}}{En_{total}} \quad (7)$$

- (d) Shannon entropy (SE): Entropy measures the degree of complexity or randomness present in the signal. The Shannon entropy is given by

$$SE = - \sum_{i=1}^{N-1} p_i \log_2 p_i x[i] \quad (8)$$

Where p_i is the probability of the i^{th} value appearing in the signal $x[n]$.

- (e) Log energy entropy (LogEnE): The other entropy measure log energy entropy is given by,

$$LogEnE = - \sum_{i=1}^{N-1} (\log_2 p_i x[i])^2 \quad (9)$$

- (f) Kurtosis (K): Kurtosis coefficient indicates sharpness of maximum value and is calculated as,

$$K = \frac{1}{N\sigma^4} \sum_{i=1}^N (x[i] - \mu^4) \quad (10)$$

- (g) Skewness (Sk): Skewness coefficient defines the symmetry disturbance of distribution which is calculated as,

$$Sk = \frac{1}{N\sigma^3} \sum_{i=1}^N (x[i] - \mu^3) \quad (11)$$

III. CLASSIFICATION

The performances of the features were evaluated through the support vector machine (SVM) [10] and K-nearest neighbor (KNN) classifiers. Both are supervised discriminative learning algorithms used for data analysis and classification. With the training dataset, SVM creates a feature space where the data of different categories are well separable. The model generates a hyperplane that maximizes the marginal distance between the categories. KNN, on the other hand, uses the minimum distance between the query instance and the neighbor samples to

determine the K nearest neighbors. The algorithm is very sensitive to the local distribution of feature vectors. Successful application of these classification models has been observed in literature for BCI based MI classification [11],[13],[14].

IV. RESULTS AND DISCUSSION

In Fig. 5 (a-g), the left hand (LH) and right hand (RH) motor imagery conditions are compared for each feature extracted from the wavelet coefficients of the EEG alpha band for C3 and C4 locations. The extracted features are averaged over the LH and RH motor imagery conditions to show the differences.

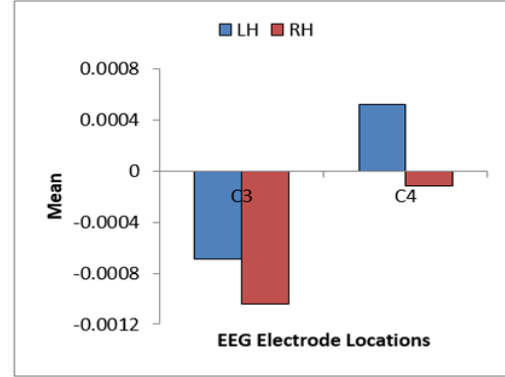


Fig 5 (a) Left Hand (LH) Vs Right Hand (RH) movement in terms of mean of alpha band

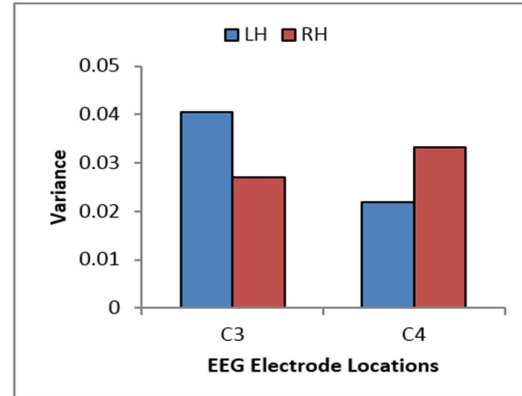


Fig 5 (b) Left Hand (LH) Vs Right Hand (RH) movement in terms of variance of alpha band

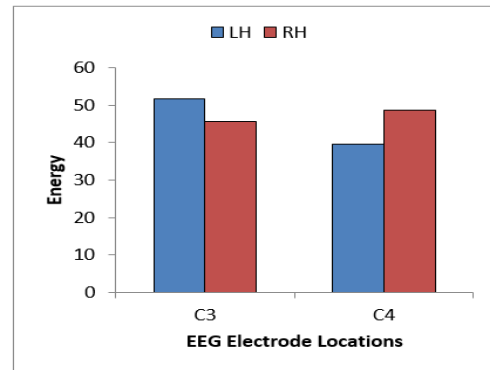


Fig 5 (c) Left Hand (LH) Vs Right Hand (RH) movement in terms of energy of alpha band

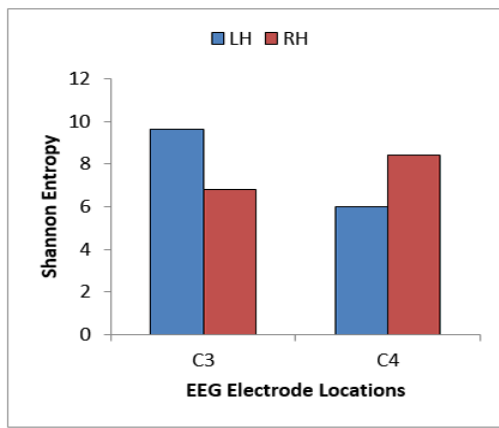


Fig 5 (d) Left Hand (LH) Vs Right Hand (RH) movement in terms of Shannon Entropy of alpha band

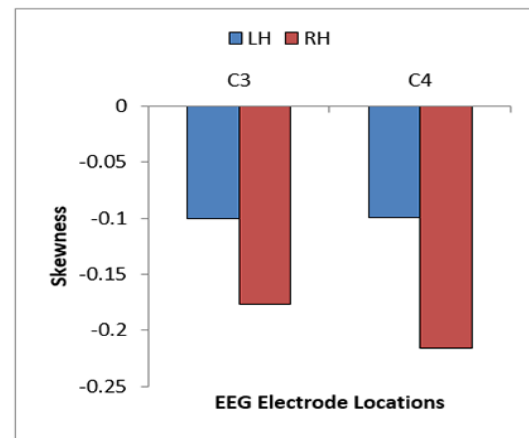


Fig 5 (g). Left Hand (LH) Vs Right Hand (RH) movement in terms of Shannon Skewness of alpha band

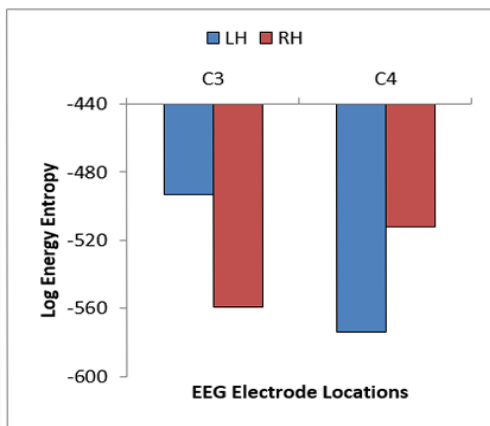


Fig 5 (e) Left Hand (LH) Vs Right Hand (RH) movement in terms of Log Energy Entropy of alpha band

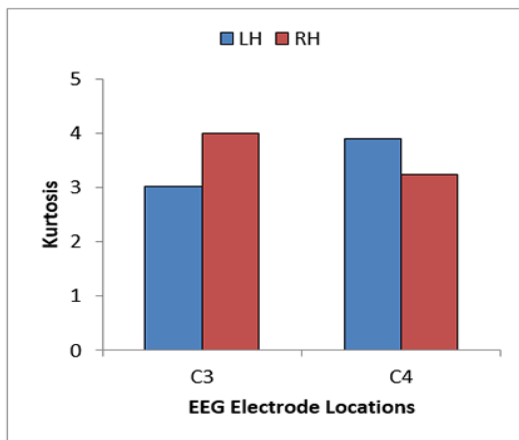


Fig 5 (f) Left Hand (LH) Vs Right Hand (RH) movement in terms of Shannon Kurtosis of alpha band

Using 10-fold cross-validation, the classification accuracies were examined using nonlinear SVM and KNN models. Table-I displays the classification accuracies as obtained with the extracted features of alpha frequency bands. The results demonstrated the effectiveness of extracted features.

Table 1. Classification Accuracies in percentage for different features of the alpha band

Features	SVM	KNN
Mean (μ)	54.3	56.4
Variance (σ^2)	84.3	82.1
Energy (En)	82.1	80.7
Shannon entropy (SE)	86.4	84.3
Log energy entropy (LogEnE)	85.0	83.6
Kurtosis (K)	83.6	82.9
Skewness (Sk)	71.4	66.4

V. CONCLUSION

In the present study, the features mentioned were extracted from the alpha frequency band obtained from DWT decomposition. The features are then fed into the SVM and KNN classifiers to differentiate the left and right limb movement. DWT is an excellent time-frequency analysis tool for non-stationary EEG signals. It provided good time as well as frequency resolution that helped in getting better classification accuracies with the extracted features. We examined seven different EEG features including mean, variance, energy, Shannon entropy, log energy entropy, Kurtosis, and skewness of the wavelet coefficients in the alpha frequency band. On examining the performances of each individual features with SVM and KNN, it is seen that the best classification accuracy is obtained with Shannon entropy feature. Being a measure of uncertainty of a system, Shannon entropy has successfully

represented the nonstationary nature of EEG signals. The results of the study produced a maximum accuracy of 86.4% which is higher than the results obtained with energy entropy (85%) in a recent study using the same data-set [14]. Further, the analysis showed superior performance of SVM over KNN in all cases.

The approach of feature extraction and classification considered in this study are very simple and robust. EEG based BCI control devices require most relevant features that can provide higher computational efficiency with lower computational time. Future study may include optimization of the feature vector and more efficient classification methodologies to implement for EEG based BCI application.

REFERENCES

- [1] W. Y. Hsu and Y. N. Sun, "EEG-based motor imagery analysis using weighted wavelet transform features," *J. Neurosci. Methods*, vol. 176, no. 2, pp. 310-318, 2009.
- [2] Herman, Pawel, Girijesh Prasad, Thomas Martin McGinnity, and Damien Coyle. "Comparative analysis of spectral approaches to feature extraction for EEG-based motor imagery classification." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 16, no. 4 (2008): 317-326.
- [3] A. Khasnobish, S. Bhattacharyya, A. Konar, and D. N. Tiwari, "K-Nearest Neighbor Classification of Left-Right Limb Movement Using EEG Data," in *International conference on Biomedical Engineering and assistive technologies*, NIT Jalandhar, 2010, pp. 1-6.
- [4] Xia, Bin, Lei Cao, Oladazimi Maysam, Jie Li, Hong Xie, Caixia Su, and Niels Birbaumer. "A binary motor imagery tasks based brain-computer interface for two-dimensional movement control." *Journal of neural engineering* 14, no. 6 (2017): 066009.
- [5] M. Teplan, "Fundamentals of EEG Measurement," *Meas. Sci. Rev.*, vol. 7, no. 2, pp. 497-503, 2008.
- [6] G. Pfurtscheller, C. Brunner, A. Schlogl, and F. H. Lopes da Silva, "Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks," *Neuroimage*, vol. 31, no. 1, pp. 153-159, May 2006.
- [7] Panda, R., P. S. Khobragade, P. D. Jambhule, S. N. Jengthe, P. R. Pal, and T. K. Gandhi. "Classification of EEG signal using wavelet transform and support vector machine for epileptic seizure detection." In *Systems in Medicine and Biology (ICSMB)*, 2010 International Conference on, pp. 405-408. IEEE, 2010.
- [8] RamaRaju, P., Anojana Aurora, N., Rao, Y.M.: Relevance of wavelet transform for taxonomy of EEG signals. In: *Electronic Computer Technology (ICECT)*, 2011 3rd International Conference on. vol. 3, pp. 466-470. IEEE (2011)
- [9] Xu, Baoguo, and Aiguo Song. "Pattern recognition of motor imagery EEG using wavelet transform." *Journal of Biomedical Science and Engineering* 1, no. 01 (2008): 64.
- [10] Guo L, Wu Y, Zhao L, Cao T, Yan W, Shen X (2011) Classification of mental task from EEG signals using immune feature weighted support vector machines. *IEEE Trans Magn* 47(5):866-869
- [11] F. Lotte, M. Congedo, A. Lecuyer, F. Lamarche, and B. Arnaldi, "A review of classification algorithms for EEG-based computer interfaces," *J. Neural Eng.*, vol. 4, pp. R1-R13, 2007.
- [12] F. Lotte, M. Congedo, A. Lecuyer, F. Lamarche, and B. Arnaldi, "A review of classification algorithms for EEG-based computer interfaces," *J. Neural Eng.*, vol. 4, pp. R1-R13, 2007.
- [13] D. Hu, W. Li, and X. Chen, "Feature extraction of motor imagery EEG signals based on wavelet packet decomposition," *2011 IEEE/ICME Int. Conf. Complex Med. Eng. C*. 2011, pp. 694-697, 2011.
- [14] R. Chatterjee, T. Bandyopadhyay, D. K. Sanyal, and D. Guha, "Comparative Analysis of Feature Extraction Techniques in Motor Imagery EEG Signal Classification," *Springer, Singapore*, 2018, pp. 73-83
- [15] M. Z. Baig, Y. Mehmood, and Y. Ayaz, "A BCI System Classification Technique Using Median Filtering and Wavelet Transform," in *Dynamics in Logistics*, Springer, Cham, 2015, pp. 355-364.